

## Phase 4

# Interview-Ready — Index & Study Checklist

Goal: convert three Phases of accumulated knowledge into genuine interview performance. This Phase opens with a repeatable six-step framework and regional panel calibration (India / UAE / Saudi Arabia), then applies that framework to fifteen fully-worked mock system-design problems — each one deliberately citing and reusing patterns from earlier Phases rather than treating every problem as new. It closes with a timed drill set and an honest inventory of common interview pitfalls. Complete chapters in order; Chapters 3 onward assume Chapter 1's framework is already internalized.

| #  | Chapter                                                                                                                                     | Time   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1  | <b>The HLD Interview Framework</b><br>Repeatable approach: requirements, estimation, API, HLD, deep dive, trade-offs                        | 1.5 hr |
| 2  | <b>India vs UAE vs Saudi Arabia Panel Expectations</b><br>Flipkart/Swiggy/Razorpay-tier vs Careem/Noon-tier vs STC/Tamara/Jahez-tier panels | 1.5 hr |
| 3  | <b>Mock Design: URL Shortener</b><br>Classic warm-up problem, full mock-interview walkthrough                                               | 1.5 hr |
| 4  | <b>Mock Design: Rate Limiter</b><br>Applying Phase 2 concepts in an interview setting                                                       | 1.5 hr |
| 5  | <b>Mock Design: A Distributed Cache</b><br>Cache eviction, consistency, scaling cache layers                                                | 1.5 hr |
| 6  | <b>Mock Design: Typeahead / Autocomplete System</b><br>Trie-based search suggestions                                                        | 1.5 hr |
| 7  | <b>Mock Design: WhatsApp / Chat System</b><br>Real-time messaging, delivery guarantees, presence, group chats                               | 2 hr   |
| 8  | <b>Mock Design: A News Feed</b><br>Fan-out on write vs read, ranking, feed generation at scale                                              | 2 hr   |
| 9  | <b>Mock Design: Twitter / X</b><br>Timeline generation, trends, celebrity fan-out problem                                                   | 2 hr   |
| 10 | <b>Mock Design: An E-Commerce Checkout</b><br>Inventory consistency, order flow, payment integration touchpoints                            | 2 hr   |
| 11 | <b>Mock Design: A Ride-Hailing / Dispatch System</b><br>Geo-indexing, matching algorithms, real-time location tracking                      | 2 hr   |

| #  | Chapter                                                                                                             | Time   |
|----|---------------------------------------------------------------------------------------------------------------------|--------|
| 12 | <b>Mock Design: A Payment Gateway</b><br>Idempotency, reconciliation, ledger design, fraud checks                   | 2 hr   |
| 13 | <b>Mock Design: A Food Delivery System</b><br>Multi-sided marketplace design, ETA prediction, order state machine   | 2 hr   |
| 14 | <b>Mock Design: A Ticket Booking System</b><br>High-concurrency seat locking (BookMyShow/IRCTC-style)               | 2 hr   |
| 15 | <b>Mock Design: A Notification System</b><br>Fan-out, delivery channels, retry/dedup logic                          | 1.5 hr |
| 16 | <b>Mock Design: A Distributed File Storage Service</b><br>Chunking, metadata service, sync conflict resolution      | 1.5 hr |
| 17 | <b>Mock Design: A Web Crawler</b><br>Crawling at scale, dedup, politeness policies                                  | 1.5 hr |
| 18 | <b>Mock Interview Drill Set &amp; Common Pitfalls</b><br>Timed practice prompts, red flags, handling "I don't know" | 2 hr   |

**Phase 4 total: ~31 hours**

## Study Checklist

Mark each chapter complete only once you can do all three of the following without notes:

- Run the chapter's mock design as a full, timed, out-loud interview using Chapter 1's six-step framework, without looking at the chapter.
- Answer all "Common Interview Questions" from memory with a full model-approach structure, naming both the pattern and its trade-off.
- Complete the practice exercise(s) unaided, and explain which earlier-Phase chapters the design leans on.

| Done?                              | Chapter                                         |
|------------------------------------|-------------------------------------------------|
| <input type="checkbox"/> Chapter 1 | The HLD Interview Framework                     |
| <input type="checkbox"/> Chapter 2 | India vs UAE vs Saudi Arabia Panel Expectations |
| <input type="checkbox"/> Chapter 3 | Mock Design: URL Shortener                      |
| <input type="checkbox"/> Chapter 4 | Mock Design: Rate Limiter                       |
| <input type="checkbox"/> Chapter 5 | Mock Design: A Distributed Cache                |
| <input type="checkbox"/> Chapter 6 | Mock Design: Typeahead / Autocomplete System    |
| <input type="checkbox"/> Chapter 7 | Mock Design: WhatsApp / Chat System             |
| <input type="checkbox"/> Chapter 8 | Mock Design: A News Feed                        |
| <input type="checkbox"/> Chapter 9 | Mock Design: Twitter / X                        |

| Done?                               | Chapter                                         |
|-------------------------------------|-------------------------------------------------|
| <input type="checkbox"/> Chapter 10 | Mock Design: An E-Commerce Checkout             |
| <input type="checkbox"/> Chapter 11 | Mock Design: A Ride-Hailing / Dispatch System   |
| <input type="checkbox"/> Chapter 12 | Mock Design: A Payment Gateway                  |
| <input type="checkbox"/> Chapter 13 | Mock Design: A Food Delivery System             |
| <input type="checkbox"/> Chapter 14 | Mock Design: A Ticket Booking System            |
| <input type="checkbox"/> Chapter 15 | Mock Design: A Notification System              |
| <input type="checkbox"/> Chapter 16 | Mock Design: A Distributed File Storage Service |
| <input type="checkbox"/> Chapter 17 | Mock Design: A Web Crawler                      |
| <input type="checkbox"/> Chapter 18 | Mock Interview Drill Set & Common Pitfalls      |

## How Phase 4 Chapters Connect

Phase 4 is deliberately ordered so complexity and cross-referencing both increase steadily:

- 1 Ch 1 gives the repeatable six-step process every later mock-design chapter runs through explicitly.
- 2 Ch 2 calibrates that process to India, UAE, and Saudi Arabia panel expectations specifically, since this program targets interviews across all three markets.
- 3 Ch 3–6 are lower-complexity warm-up-tier problems (URL shortener, rate limiter, cache, typeahead), each revisiting one Phase 1/2/3 concept in interview form.
- 4 Ch 7–9 build real-time and feed-shaped systems (chat, news feed, Twitter), introducing and then reusing the fan-out-on-write vs. fan-out-on-read tension across three consecutive chapters.
- 5 Ch 10–12 build correctness-critical transactional systems (checkout, ride-hailing, payments), centered on atomic updates, Sagas, and ledger design.
- 6 Ch 13–14 are genuine combination problems (food delivery, ticket booking) that deliberately require citing multiple earlier Phase 4 chapters at once.
- 7 Ch 15–17 close out supporting/infrastructure systems (notifications, file storage, web crawler) that earlier chapters referenced as dependencies without designing in full.
- 8 Ch 18 withholds worked answers entirely, handing you fresh drills and an honest pitfalls inventory — the final bridge from studied material to real interview readiness.

### Program complete

Once every chapter is checked off and Chapter 18's fresh drills have each been run as a full timed mock interview, the complete HLD Learning + Interview-Prep Program — all four Phases, covering fundamentals through interview-ready mock designs — is finished. Revisit any chapter across any Phase as needed before real interviews; the cross-referencing built throughout this program means no chapter is truly isolated from the rest.